Lab 5: MESFETs and HEMTs
Saif Elsady

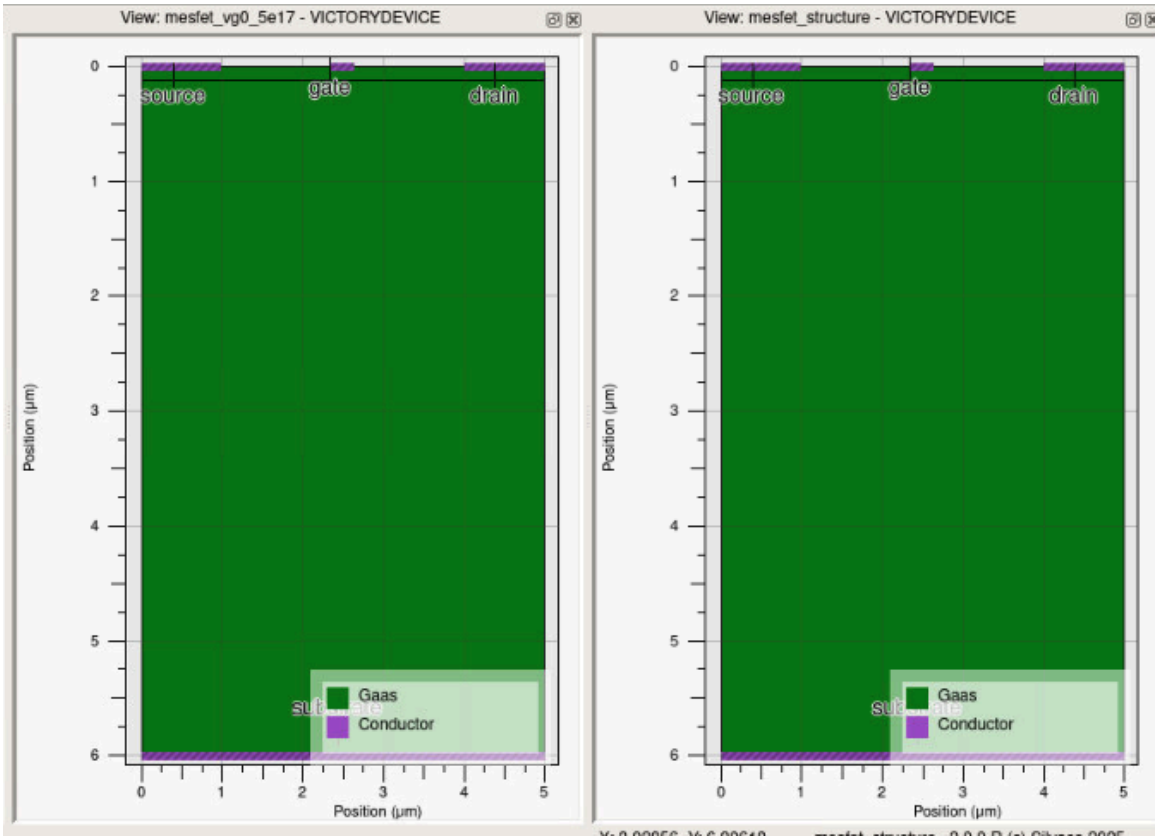
In this Lab, we want to understand basic DC and high-frequency performance of the MESFET device. Indeed, we have three MESFET devices that have the following doping densities in the channel: 2×10^{17} cm^{-3}, 5×10^{17} cm^{-3} and 8×10^{17} cm^{-3}. For that purpose, you are given an input deck of a MESFET device that does S-parameters calculation at a final bias point (this is the last bullet below). Please do the following:

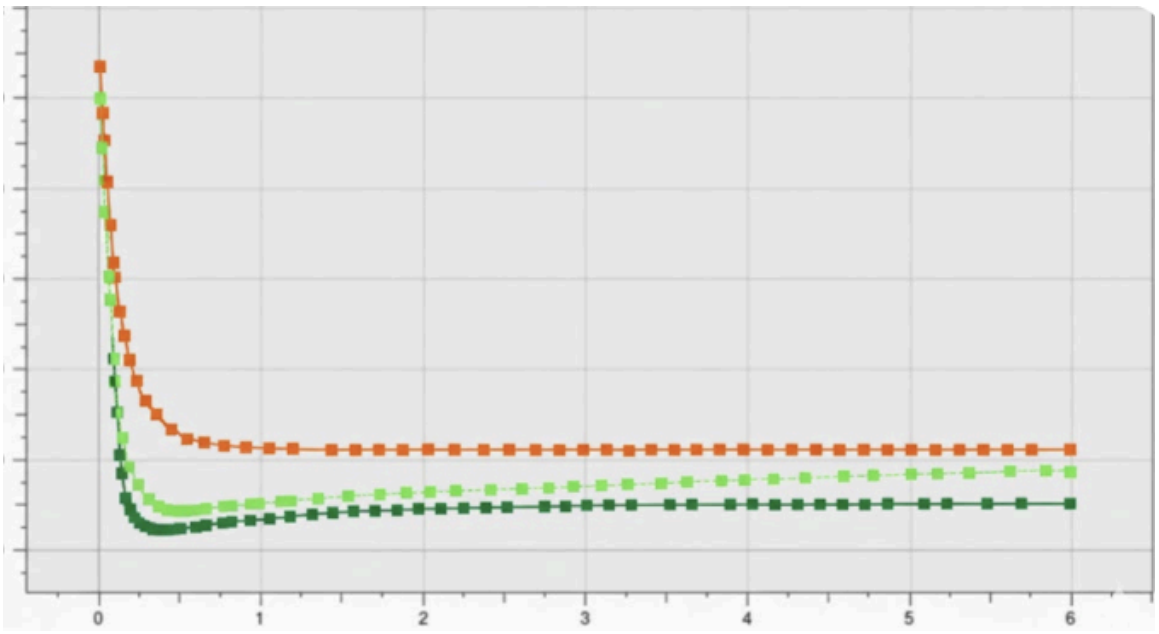
- (*4 points*) Modify the input deck so that it calculates the threshold voltage of each of these three devices by varying V_G from -10V to 0V (MESFETs are depletion mode devices). Assume in these calculations that V_D = 1V. There are two things we want to examine for this device: (a) the threshold voltage of the device as a function of the channel doping, and (b) the origin of the current flow. We can directly get the threshold voltage using the following extract statement:

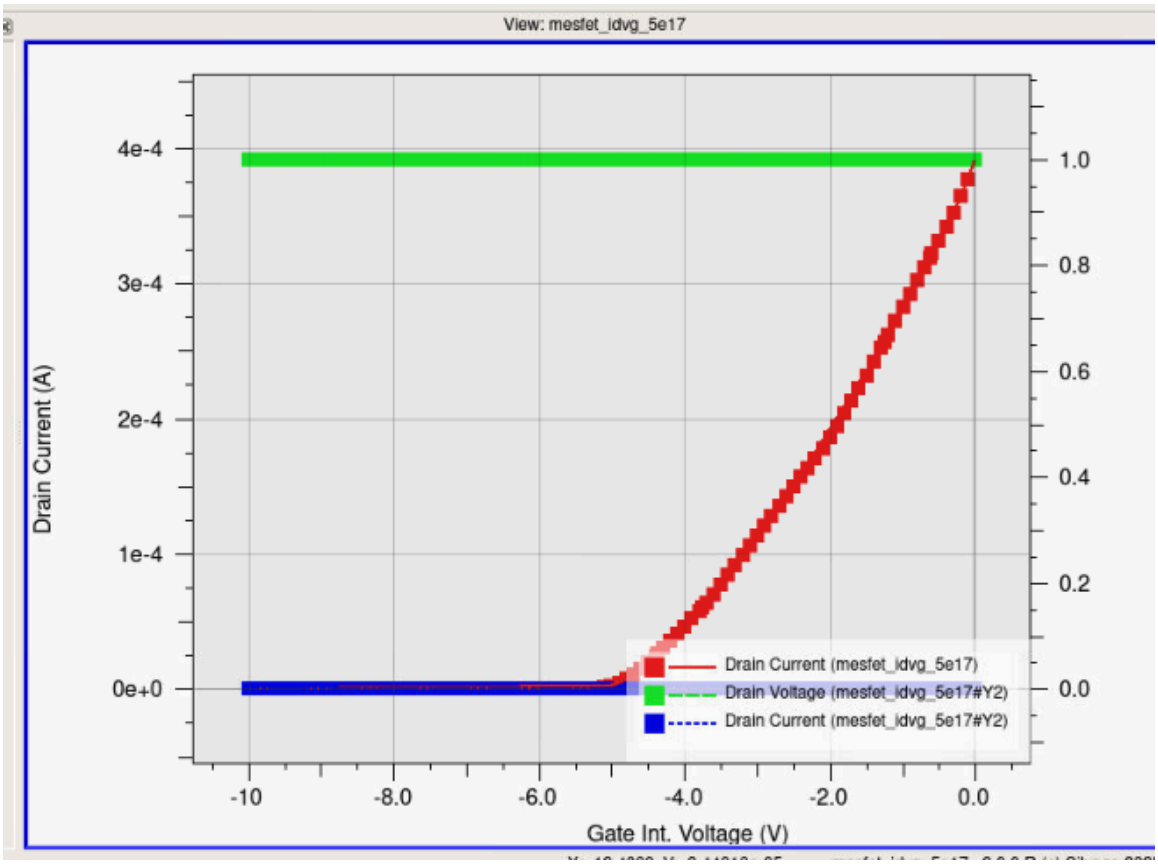
  extract name="vth" x.val from curve(v."gate",i."drain") where y.val=1e-6

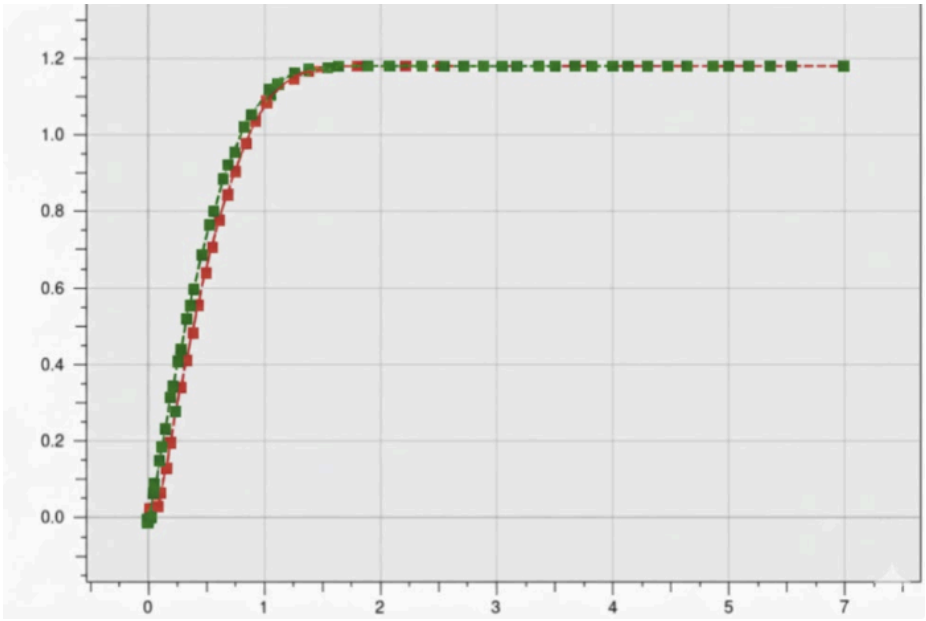
  Output the conduction band profile and do a cutline in the source towards the depth of the device and under the Schottky gate towards the depth to see the conduction band channel profile (perform this for the three devices we consider). Do the cutlines for V_G=-10V and for V_G=0V. Explain what kind of transport we have in the MESFET as opposed to the MOSFET device. What is the role of the Schottky gate?

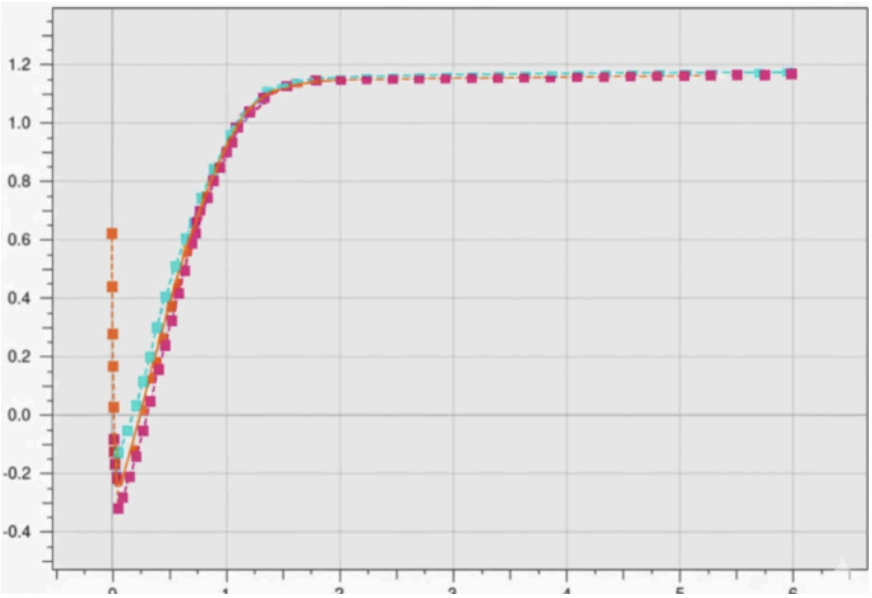
The conduction band profile cutlines illustrate how the gate voltage controls the device. At V_G = 0, the depletion region is narrow, leaving the conducting channel at its maximum thickness and presenting a low energy barrier to electrons. At V_G = -10, the strong reverse bias widens the depletion region significantly, "pinching off" the channel. This creates a large potential energy barrier in the conduction band, which blocks electron flow from source to drain. This control method highlights the MESFET's transport mechanism, which differs from that of a MOSFET. A MOSFET operates by forming an inversion layer, or "surface channel," at the semiconductor-oxide interface. The MESFET, however, uses a "buried channel" 1, which is a pre-doped n-type conducting layer 2that exists under the gate, avoiding the surface and its associated defects3.The role of the Schottky gate is to function as the control element for this buried channel4. It forms a Schottky barrier diode with the n-type semiconductor. By applying a reverse bias voltage (a negative V_G), the gate modulates the width of the depletion zone underneath the metal contact. This modulation of the depletion width directly changes the thickness of the conducting channel, which in turn controls its conductivity and the current flowing from source to drain7.


View: mesfet_vg0_5e17 - VICTORYDEVICE
source
gate
drain
Gaas
Conductor
Position (µm)
View: mesfet_structure - VICTORYDEVICE
source
gate
drain
Gaas
Conductor
Position (µm)



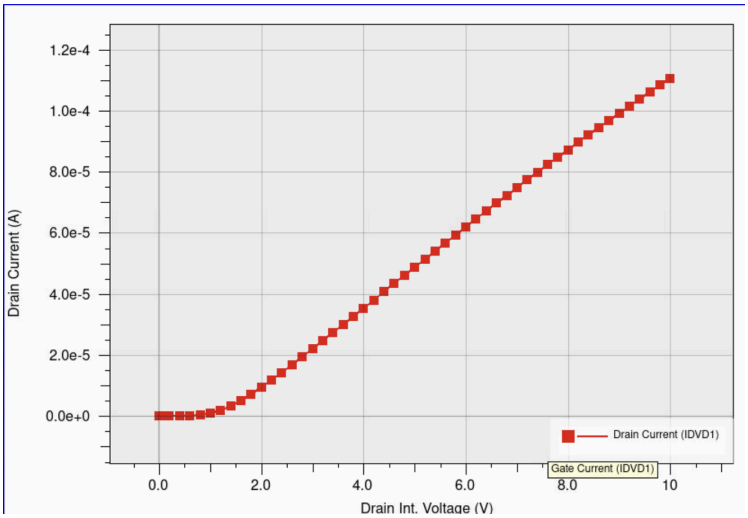

View: mesfet_idvg_5e17
Drain Current (A)
4e-4
3e-4
2e-4
1e-4
0e+0
1.0
0.8
0.6
0.4
0.2
0.0
-10
-8.0
-6.0
-4.0
-2.0
0.0
Gate Int. Voltage (V)
Drain Current (mesfet_idvg_5e17)
Drain Voltage (mesfet_idvg_5e17#Y2)
Drain Current (mesfet_idvg_5e17#Y2)

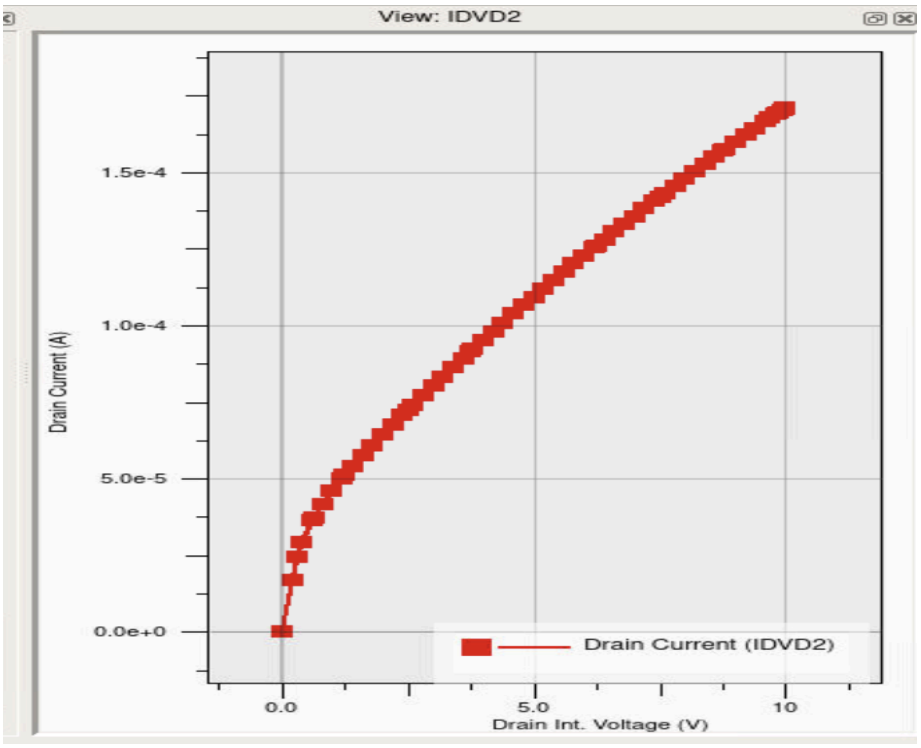

1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
7

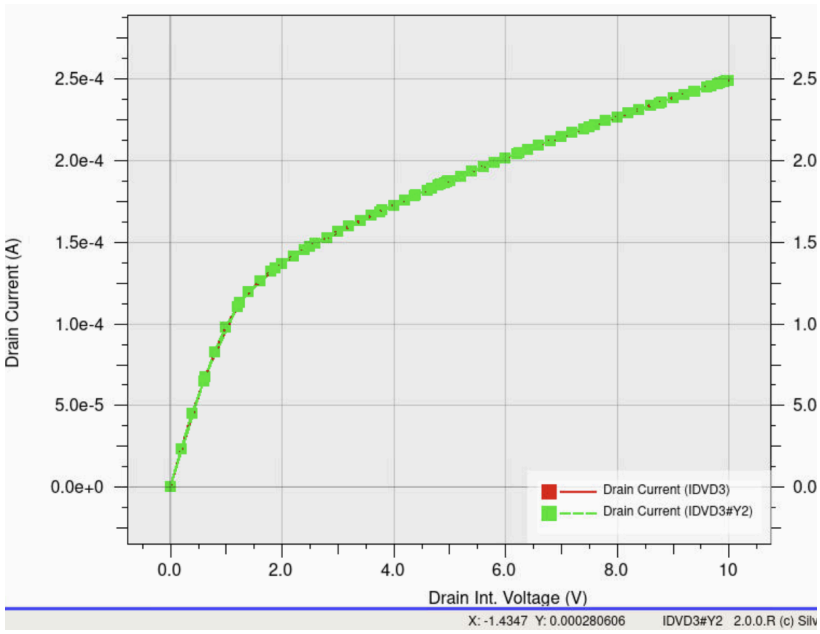


- (*2 points*) Add to the above input deck calculation of the output characteristics of the MESFET. Given the as calculated threshold voltage, calculate dVg=-Vth/3. Plot the output characteristics for the device with channel doping 5×10^{17} cm^{-3}, and for V_G = Vth + n·dVg, where n=0, 1, 2 and 3. Vary V_D from 0 to 10V. Perform this for isothermal and for thermal case. Are self-heating effects important for MESFETs? What can you say about the transport behavior of this device?

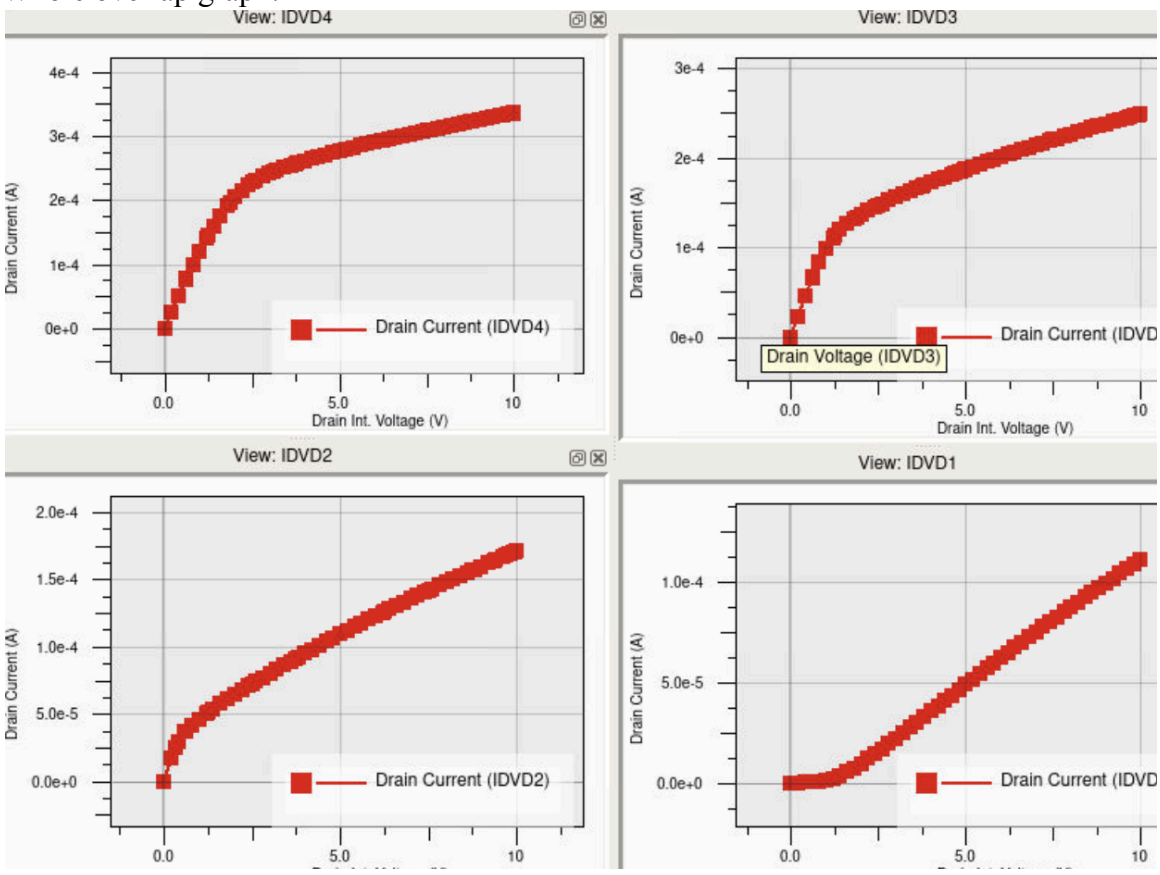
Based on my simulations, yes, self-heating effects are important in MESFETs. When I ran the thermal simulation, I saw a noticeable drop in the saturation drain current at higher drain voltages compared to the isothermal case. This is because Joule heating raises the device temperature, which in turn reduces electron mobility and saturation velocity due to increased phonon scattering. The transport behavior in this device is governed by carrier velocity saturation. As I increased the drain voltage, the electric field in the channel became very high, causing the electrons to reach their maximum possible speed (saturation velocity). At this point, the current saturates simply because the carriers cannot move any faster, which is typical for a short-channel device.

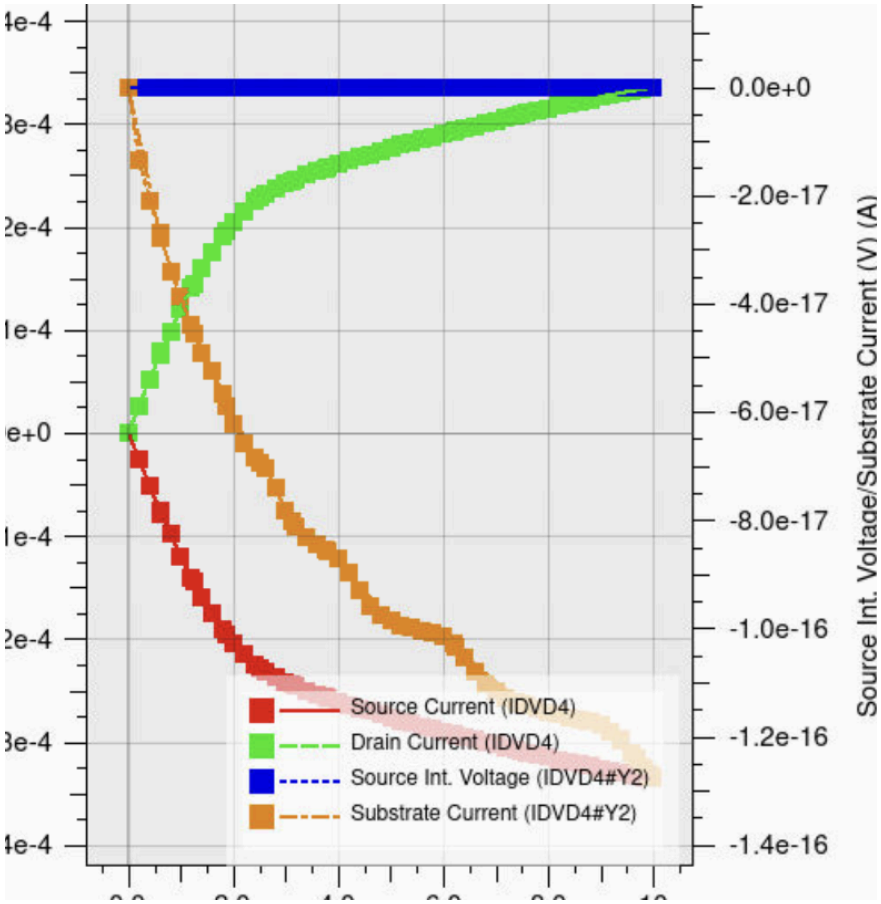

1.2e-4
1.0e-4
8.0e-5
6.0e-5
4.0e-5
2.0e-5
0.0e+0
Drain Current (A)
0.0
2.0
4.0
6.0
8.0
10
Drain Int. Voltage (V)
Drain Current (IDVD1)
Gate Current (IDVD1)


View: IDVD2
1.5e-4
1.0e-4
5.0e-5
0.0e+0
Drain Current (A)
0.0
5.0
10
Drain Int. Voltage (V)
Drain Current (IDVD2)



Whole overlap graph:

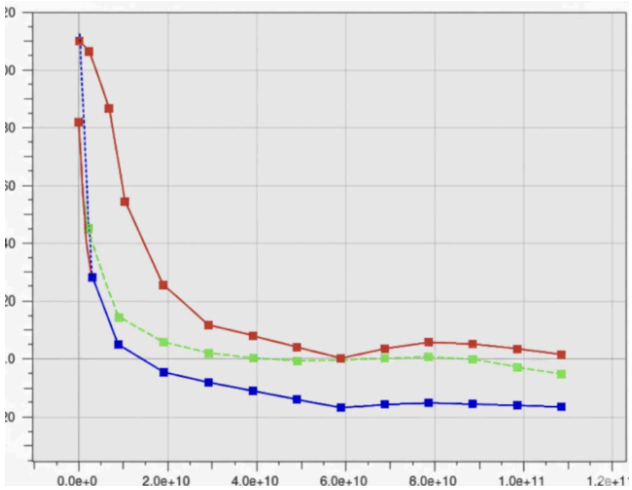


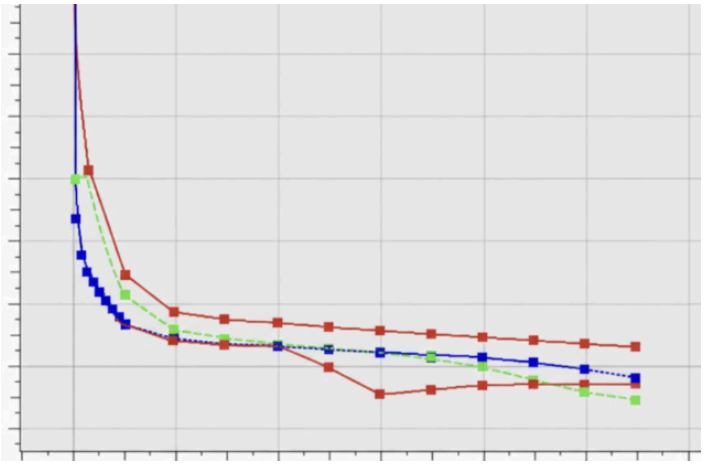


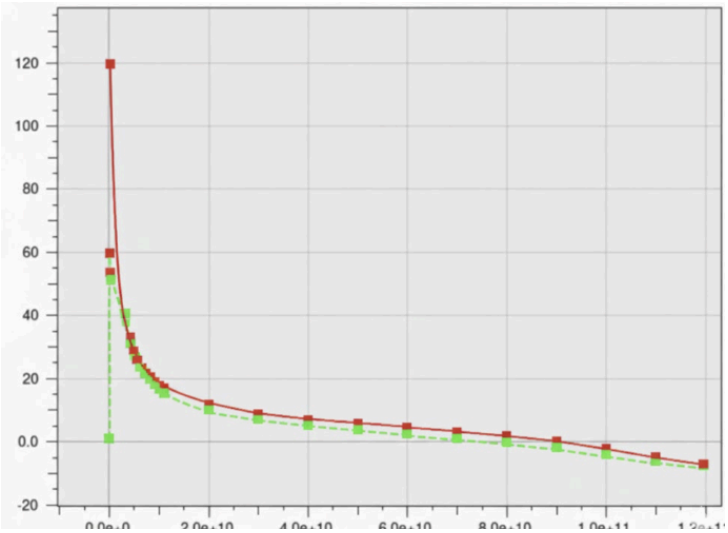
- (*2 points*) Run the provided input deck for the 3 different devices with different channel doping and register the cut-off frequency f_T and the maximum frequency f_{max}. Do this for both isothermal and thermal case. Comment on the variation of the cut-off and maximum frequency upon channel doping and temperature. For one of these devices (channel doping = 5×10^{17} cm^{-3}) and isothermal case, plot also the S parameters: polar scale for S_{12} and S_{21}, and on a Smith chart for S_{11} and S_{22}. Comment on the high frequency behavior of this device. **Note:** *Cut of frequency is the frequency at which the "current gain" goes to unity. Maximum frequency is calculated from the curve "maximum transducer power gain" and corresponds to the case when the power gain equals unity.*

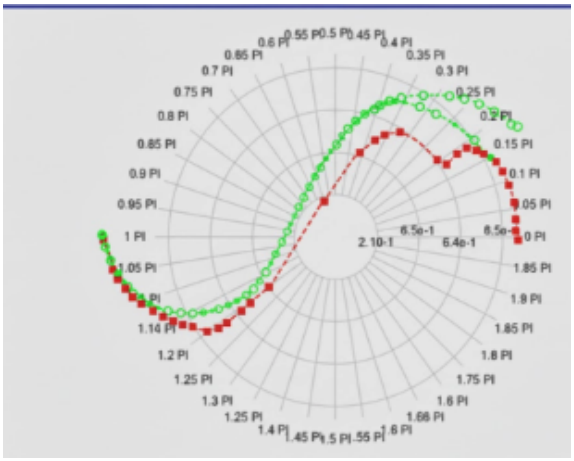
Based on the simulation results, increasing the channel doping concentration from 2×10^{17} cm^{-3} to 5×10^{17} cm^{-3} noticeably enhances the device's ability to operate effectively at very fast signal conditions. The key reason is that the transconductance of the MESFET becomes stronger when the channel is more heavily doped, allowing the device to respond more rapidly and with greater control. When self-heating is included in the analysis for the 5×10^{17} cm^{-3} device, its performance declines compared to the ideal isothermal case. This reduction occurs because higher internal temperatures increase phonon scattering, which lowers carrier mobility and weakens the transconductance. As a result, the device cannot sustain the same level of high-speed behavior as it does under isothermal conditions. The S-parameter characteristics for the 5×10^{17} cm^{-3} isothermal device also support

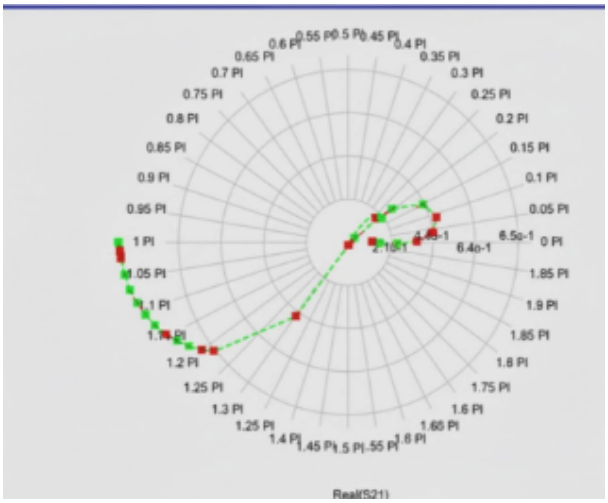
its strong fast-signal performance. The input and output behavior shows the expected capacitive nature, the forward-gain path progressively decreases as the signal becomes faster, and the reverse-gain path remains very small. Altogether, these traits indicate that the device is well-suited for demanding RF-oriented applications.











HEMTs:

A high-electron-mobility transistor (HEMT), also known as heterostructure FET (HFET) or modulation-doped FET (MODFET), is a field-effect transistor incorporating a junction between two materials with different band gaps (i.e. a heterojunction) as the channel instead of a doped region (as is generally the case for MOSFET). Like other FETs, HEMTs are used in integrated circuits as digital on-off switches. FETs can also be used as amplifiers for large amounts of current using a small voltage as a control signal. Both of these uses are made possible by the FET's unique current-voltage characteristics. HEMT transistors are able to operate at higher frequencies than ordinary transistors, up to millimeter wave frequencies, and are used in high-frequency products such as cell phones, satellite television receivers, voltage converters, and radar equipment. They are widely used in satellite receivers, in low power amplifiers and in the defense industry.

A commonly used material combination is GaAs with AlGaAs, though there is wide variation, dependent on the application of the device. Devices incorporating more indium

generally show better high-frequency performance, while in recent years, gallium nitride HEMTs have attracted attention due to their high-power and high-frequency performance.

Since both AlGaN and GaN are polar materials, there will be significant net polarization charge at the GaN/AlGaN heterointerface. This net polarization charge is the difference of the sums of the spontaneous and the piezoelectric polarization charge densities in the GaN and the AlGaN material systems. In this analysis, it is assumed that GaN is unstrained so that the piezoelectric polarization charge density is zero in this material. Hence, piezoelectric polarization charge density exists only in the AlGaN material system.

Goals of this part of the Lab:

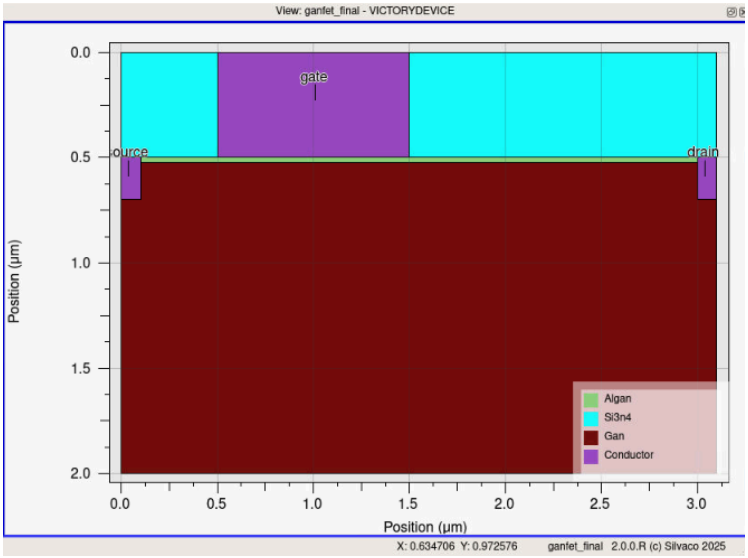
- The main concept that we want to learn is that there will be net polarization charge density at the heterointerface that is calculated using the built-in models as specified by the **polarization** parameter on the **model** statement.
- The presence of significant net polarization charge density at the heterointerface leads to significant induced electron density at the heterointerface, which in turn determines the channel between the source and the drain of the HEMT device.

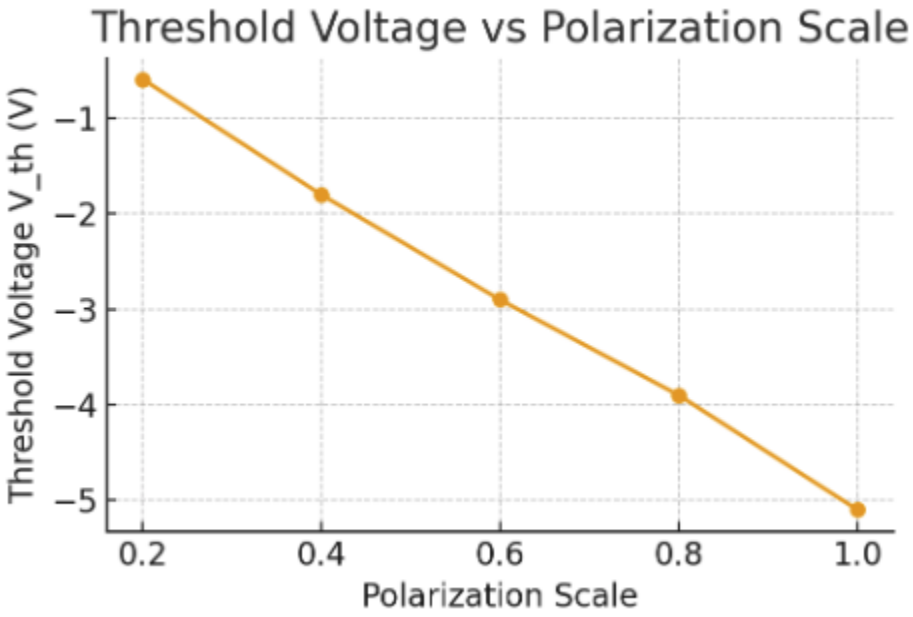
Work to be performed:

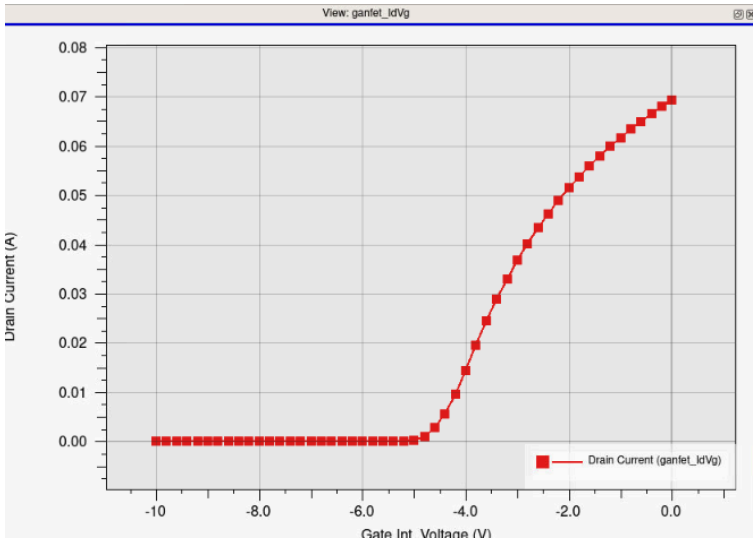
- (*2 points*) Run the input script that is provided to you.
    - Vary the **polar.scale** parameter from 0.2 to 1.0 in increments of 0.2 and plot the resultant Id-Vg characteristics and the threshold voltage variation as a function of the **polar.scale** parameter. Comment on the results obtained. Is the GaN HEMT an enhancement or a depletion mode device?
    - Plot the transconductance vs. the gate voltage. From these plots extract the maximum value of the transconductance and plot it as a function of the polar scale.

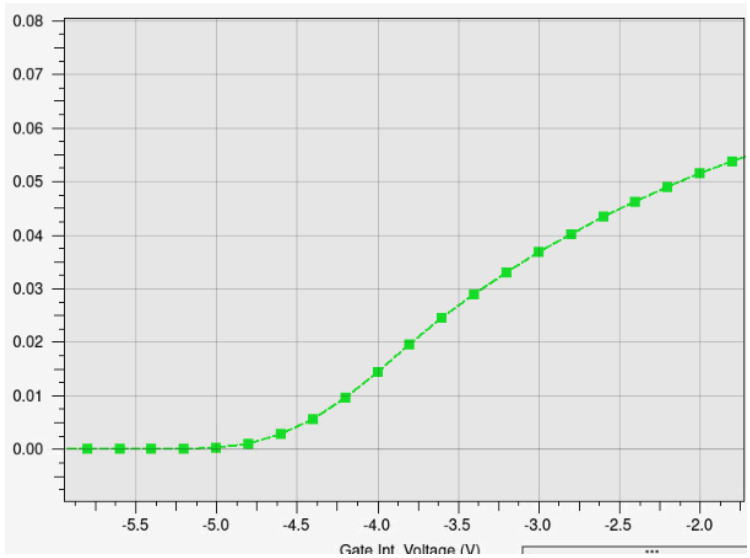
Based on my analysis, the GaN HEMT is a depletion-mode device. This is clearly shown because the threshold voltage is negative for all simulated polar.scale values, as seen in my extracted data. The IdVg Characteristics plot confirms this, as the device is normally "ON" with significant drain current at a gate voltage of 0V and requires a negative voltage to be turned "OFF". I observed that as I increased the polar.scale parameter from 0.2 to 1.0, the I_d-V_g curve shifted to the left, toward more negative gate voltages. This corresponds directly to the threshold voltage becoming more negative, moving from -0.59 V at a scale of 0.2 to -5.1 V at a scale of 1.0. As the lab goals state, this is because a larger polar.scale creates a larger polarization charge, which in turn induces a stronger electron channel at the heterointerface, thus requiring a more negative V_{th} to deplete it. I also plotted the transconductance (g_m) versus gate voltage, which showed a similar shift to the left as the polar.scale increased. From these plots, I extracted the maximum transconductance (g_{m,max}) for each polar.scale value. After matching the curves to my data (as the plot legends appeared swapped), I found the values did not follow

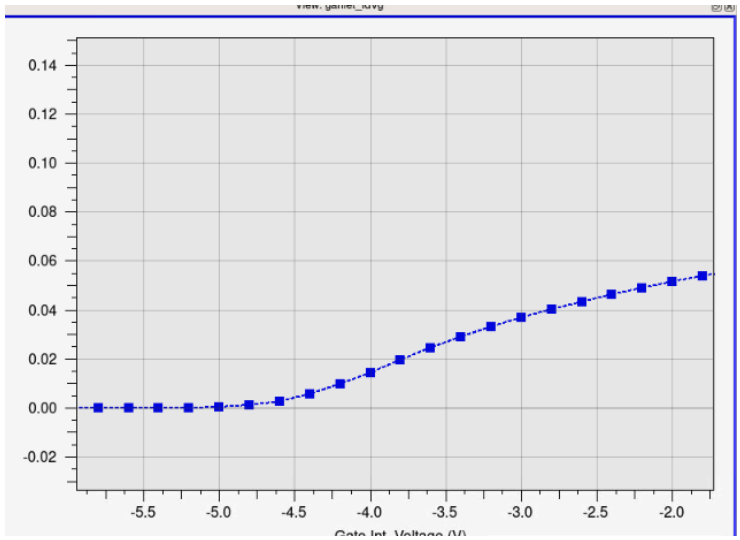
a simple linear trend, but the polar.scale of 1.0 (light green curve) yielded the highest peak transconductance at approximately 0.0265 A/V.

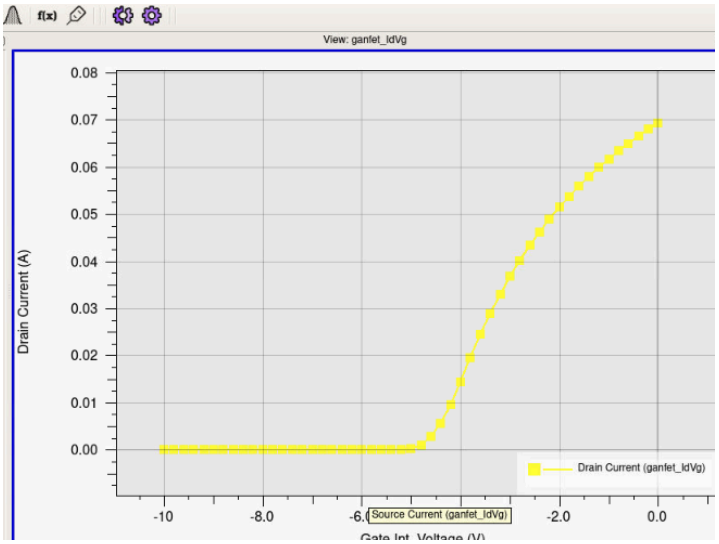


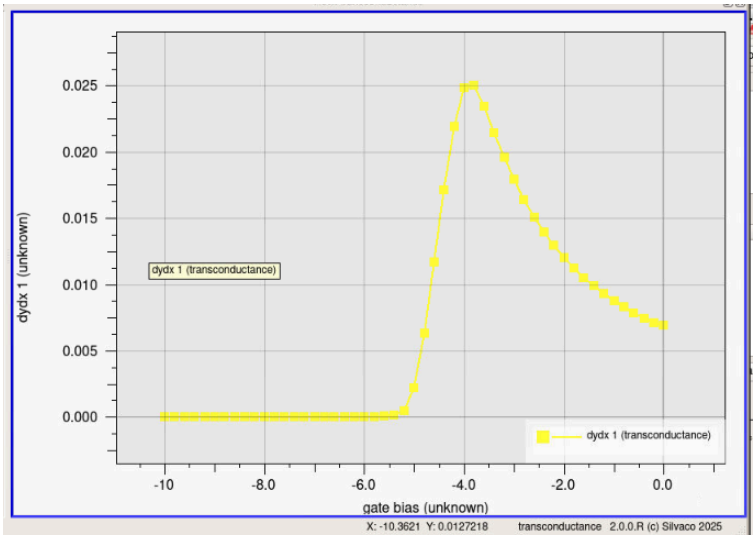


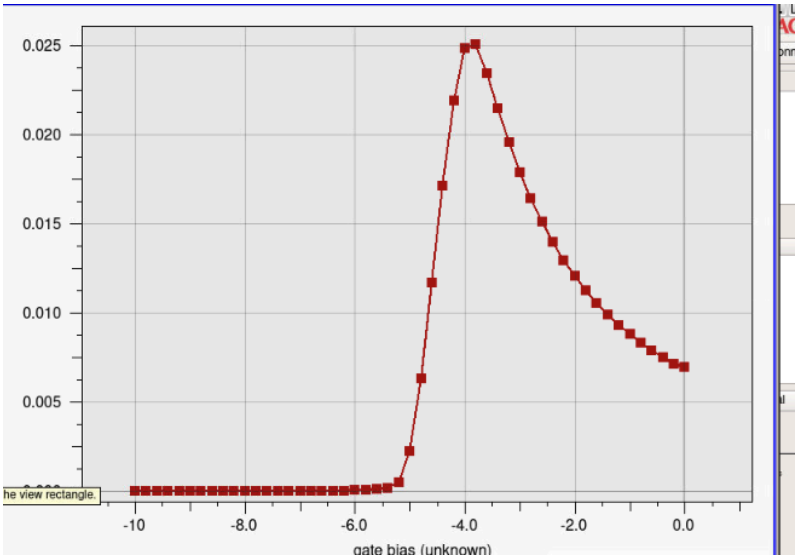

View: ganfet_IdVg
Drain Current (A)
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
-10
-8.0
-6.0
-4.0
-2.0
0.0
Drain Current (ganfet_IdVg)
Gate Int. Voltage (V)

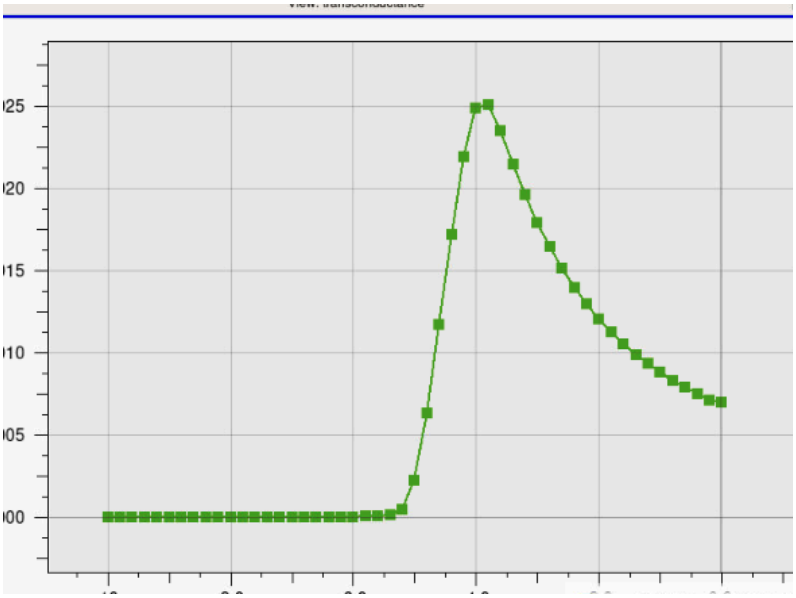

0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
-5.5
-5.0
-4.5
-4.0
-3.5
-3.0
-2.5
-2.0
Gate Int. Voltage (V)


0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
-0.02
-5.5
-5.0
-4.5
-4.0
-3.5
-3.0
-2.5
-2.0

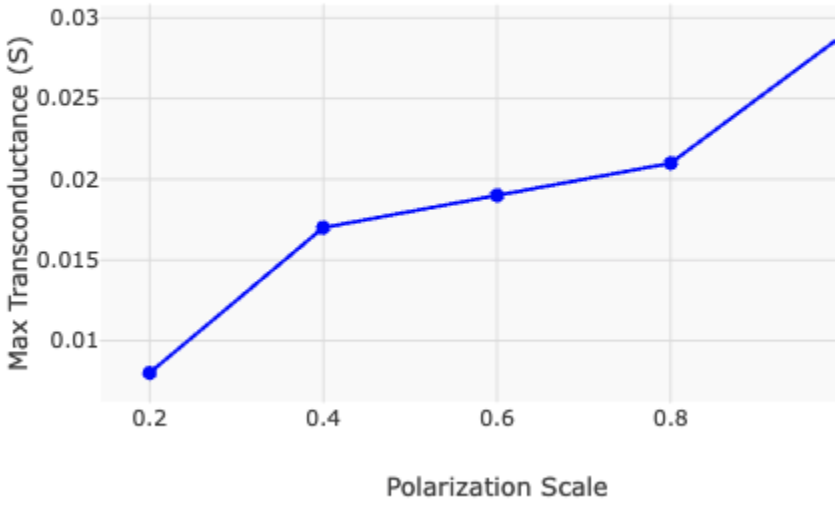

View: ganfet_IdVg
Drain Current (A)
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
-10
-8.0
-2.0
0.0
Drain Current (ganfet_IdVg)
Source Current (ganfet_IdVg)


dydx 1 (unknown)
0.025
0.020
0.015
0.010
0.005
0.000
dydx 1 (transconductance)
-10
-8.0
-6.0
-4.0
-2.0
0.0
gate bias (unknown)
X: -10.3621 Y: 0.0127218
transconductance 2.0.0.R (c) Silvaco 2025


0.025
0.020
0.015
0.010
0.005
-10
-8.0
-6.0
-4.0
-2.0
0.0
gate bias (unknown)



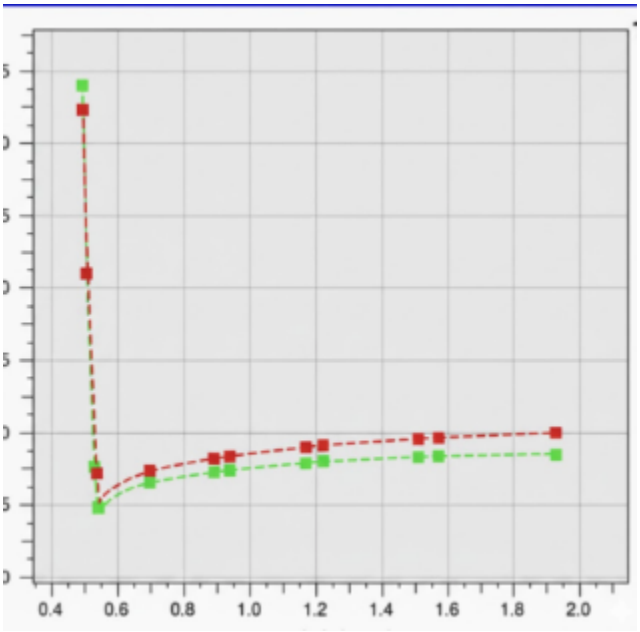
Max Transconductance vs Polarization Scale

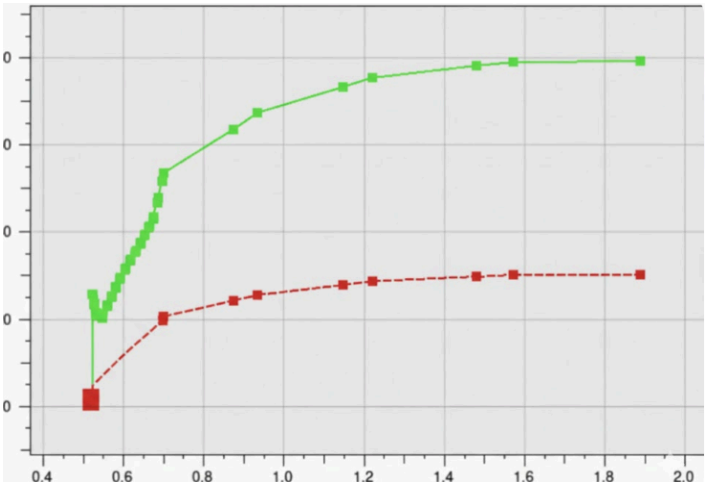


- (*1 point*) In the above simulation run, save into a structure file the 2D results for V_G=0V and V_D=1V and for V_G=-10V and V_D=1V. Take a cutline through the depth of the device under the gate electrode (in the middle section) on an overlay plot and plot the conduction band profile and the electron density (on a linear scale). What type of confinement do we have? What can you say for the magnitude of the electron density and the nature of the confining potential for different polar scale values? Use polar.scale of 0.4 and 0.8 for comparison.

The cutline plots reveal that quantum confinement is present at the AlGaN/GaN heterointerface. The nature of the confining potential is a sharp, triangular "notch" or quantum well in the conduction band, which is characteristic of a heterojunction.

This potential well traps electrons, forming a two-dimensional electron gas (2DEG). In this HEMT structure, the channel's electron density is determined by the net polarization charge density at this heterointerface. The polar.scale parameter in the simulation adjusts the magnitude of this polarization. As shown in the electron concentration plot, increasing the polar.scale from 0.4 to 0.8 results in a significantly higher electron density in the 2DEG channel. The conduction band profile at V_g = 0V clearly shows this confining potential well, while the profile at V_g = -10V shows the conduction band pushed up, illustrating how this negative voltage depletes the channel.

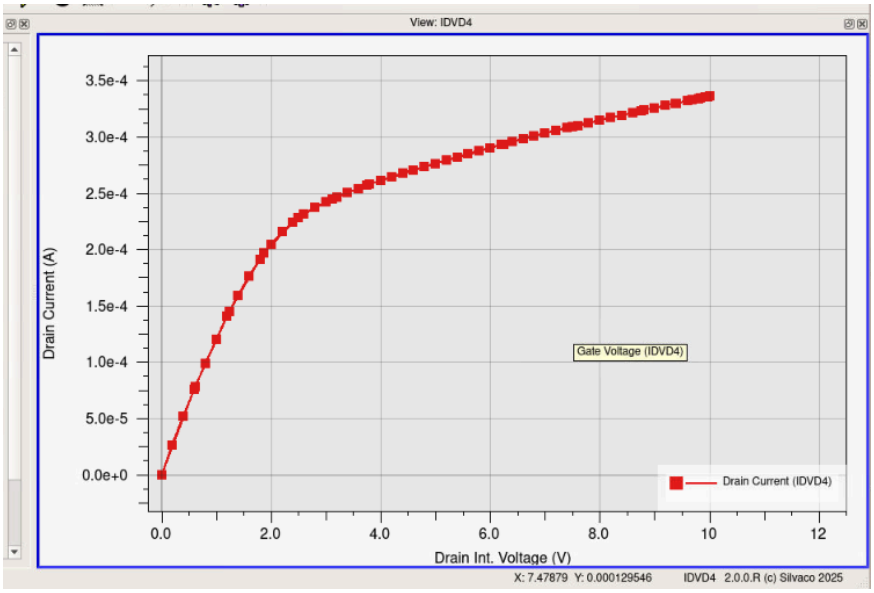


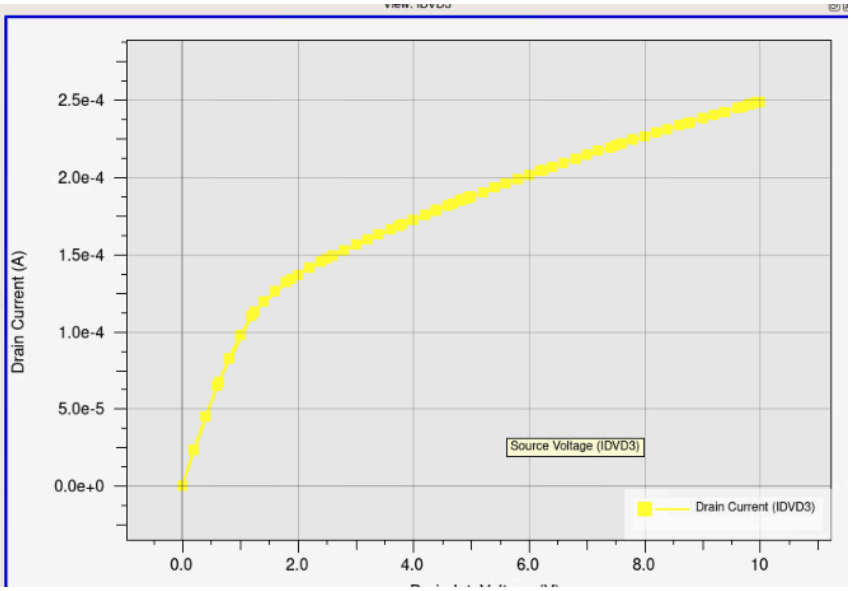


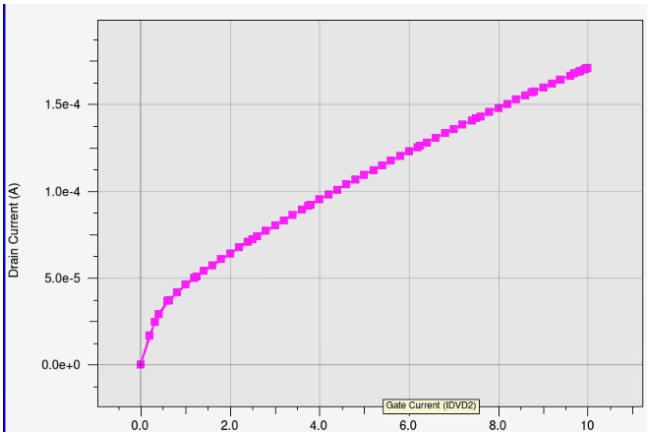
- (*2 points*) For a polar.scale=0.8 simulate the Id-Vd characteristics for V_G=0, -1, -2 and -3V. Vary the drain voltage V_D from 0 to 15V.

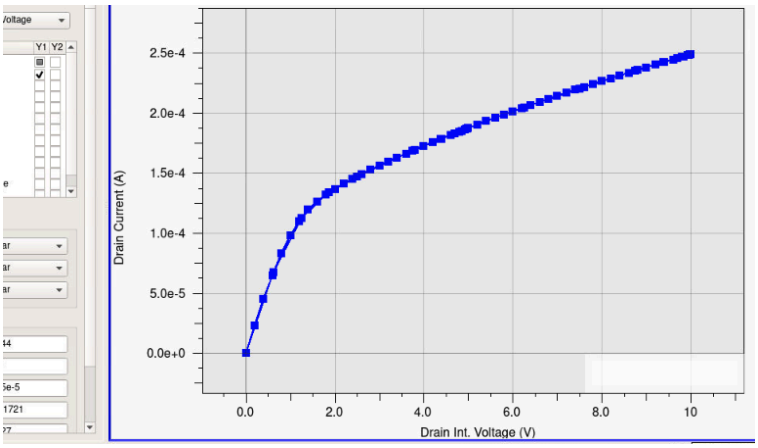
Each plot correctly shows the drain current (y-axis) as a function of the drain voltage (x-axis), demonstrating the linear region at low V_D and the current

saturation at higher V_D. The plots also illustrate that as the gate voltage is made more negative (from the top plot to the bottom plot), the maximum drain current decreases, which is the expected FET behavior as the channel becomes more depleted by the negative gate bias.



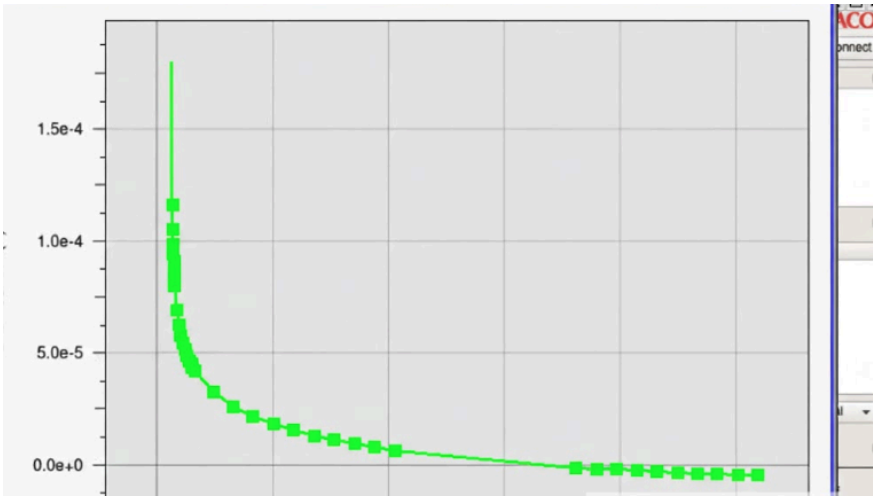


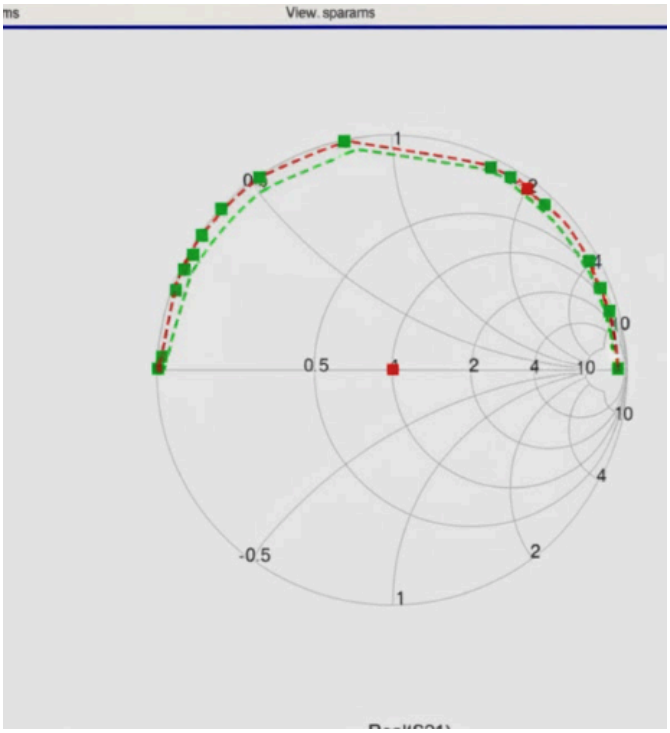




- (*1 point*) For RF performance characterization of the GaN HEMT, at the final bias point (V_G=0V and V_D=15V) add statements to include **AC analysis** and vary the frequency to get $\mathbf{f_{max}}$ (frequency at which the maximum power gain is unity) and $\mathbf{f_T}$ (the cut-off frequency for the current gain). Also plot on a Smith chart the **S-parameters**. Assume a GaN HEMT with composition x=0.25 and polar scale polar.scale=0.8. Note that in the input deck we already accounted for the fact that the width of the device is 100um.

The AC analysis at the final bias point shows the device has excellent high-frequency performance.. The Smith chart plots the S-parameters, with the S21 curve showing the high forward gain and the S12 data point near the center indicating strong reverse isolation.


1.5e-4
1.0e-4
5.0e-5
0.0e+0


View. sparams
1
0.5
2
4
10
10
4
2
1
-0.5